

# The IMAT Syllabus

The official Ministry programme (Allegato A), translated in full and laid out the IMAT.club way - every section, exactly as published.

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**Source: Decreto Ministeriale (MUR), Allegato A**

Official syllabus programme

**imat.club**

Club AI, Pro-only

This is our own English translation of the official Italian syllabus, offered as guidance only and not official or legal advice. The Ministry's original decree is the binding source in case of any discrepancy, always verify current details yourself.

# Format & scoring

The syllabus on the following pages is what these 60 questions are drawn from, current CINECA/University format.

|                      |                      |
|----------------------|----------------------|
| Total questions      | 60                   |
| Time allowed         | 100 minutes          |
| Options per question | 5 (A–E), one correct |
| Correct answer       | +1.5                 |
| Wrong answer         | –0.4                 |
| Unanswered           | 0                    |
| Maximum score        | 90                   |

## Section breakdown

|                                                |              |
|------------------------------------------------|--------------|
| Reading Skills & Knowledge Acquired in Studies | 4 questions  |
| Logical Reasoning & Problems                   | 5 questions  |
| Biology                                        | 23 questions |
| Chemistry                                      | 15 questions |
| Physics & Mathematics                          | 13 questions |

Per the decree's own preamble: candidates are expected to understand and analyze written texts of various kinds, carry out logical-mathematical reasoning, and hold general knowledge (especially historical, geographic, social, and institutional) plus subject knowledge in mathematics, chemistry, physics, and biology, at the level promoted by Italian secondary schools' national curricula.

## SECTION 1

# Reading Skills & Knowledge Acquired in Studies

The ability to understand written texts in English of different kinds and with different communicative purposes is a cross-cutting competency, since all question types are formulated in English, including through symbolic language.

The following abilities are also specifically tested: understanding abstract, uncommon, or specialized vocabulary in real contexts; identifying textual cohesion and coherence; and extracting and inferring specific information from the text. These are tested using short passages from scientific essays, classic and contemporary fiction, or current-affairs texts from newspapers and general or specialized magazines.

Also from short texts of varied type and theme, questions test knowledge acquired in prior studies and general knowledge, including at a supranational level or on topics of contemporary public debate. In particular, questions aim to assess:

- the ability to place historical-cultural phenomena in space and time;
- knowledge of the main national and international institutions;
- understanding of phenomena related to legal, economic, and citizenship matters.

## SECTION 2

# Logical Reasoning & Problems

Questions test the ability to logically complete a line of reasoning, consistent with given premises. These premises are stated in symbolic or verbal form, and concern cases or problems - including abstract ones - whose solution requires different forms of logical reasoning.

# Biology

- The chemistry of living things.
- The biological importance of weak interactions.
- Organic molecules present in organisms and their functions. The role of enzymes.
- The cell as the basis of life. Cell theory. Cell size. Prokaryotic and eukaryotic cells, animal and plant. Viruses.
- The cell membrane: structure and functions; transport across the membrane. Cell structures and their specific functions.
- Cell cycle and cell reproduction: mitosis and meiosis - chromosome complement and chromosome maps.
- Reproduction and heredity. Life cycles. Sexual and asexual reproduction.
- **Genetics:**  
Mendelian genetics (Mendel's laws and applications); classical genetics (chromosomal theory of heredity, inheritance patterns); molecular genetics (DNA structure and replication, the genetic code, protein synthesis, prokaryotic DNA, eukaryotic chromosome structure, genes and gene expression regulation); human genetics (transmission of mono- and polyfactorial traits, autosomal and X-linked hereditary diseases).
- Mutations. Natural and artificial selection. Theories of evolution. Genetic basis of evolution. Heredity and environment.
- Biotechnology: recombinant DNA technology and its applications.
- Anatomy and physiology of animals and humans. Animal tissues. Anatomy and physiology of human systems and their interactions. Homeostasis.
- Bioenergetics. The energy currency of cells: ATP. Oxidation-reduction reactions in living things. Energy processes: photosynthesis, glycolysis, aerobic respiration, and fermentation.

# Chemistry

- The constitution of matter: states of matter; heterogeneous and homogeneous systems; compounds and elements.
- Laws of ideal gases.
- Atomic structure: elementary particles; atomic number and mass number, isotopes, electronic structure of atoms of various elements.
- The periodic table: groups and periods; transition elements. Periodic properties: atomic radius, ionization potential, electron affinity, metallic character. Relationships between electronic structure, position in the periodic table, and element properties.
- Chemical bonding: ionic, covalent, and metallic bonds. Bond energy. Bond polarity. Electronegativity. Intermolecular bonds.
- Fundamentals of inorganic chemistry: nomenclature and main properties of oxides, hydroxides, acids, salts.
- Chemical reactions and stoichiometry: atomic and molecular mass, Avogadro's number, the mole concept and its application, elementary stoichiometric calculations, balancing simple reactions, different reaction types.
- Solutions: solvent properties of water, solubility, the main ways of expressing concentration.
- Equilibria in aqueous solution.
- Elements of chemical kinetics and catalysis.
- Oxidation and reduction: oxidation number, oxidizing/reducing agents. Balancing simple reactions.
- Acids and bases: the concept of acid and base. Acidity, neutrality, and basicity of aqueous solutions. pH. Hydrolysis. Buffer solutions.
- Fundamentals of organic chemistry: bonds between carbon atoms, molecular and structural formulas, isomerism. Aliphatic, alicyclic, and aromatic hydrocarbons. Functional groups: alcohols, ethers, amines, aldehydes, ketones, carboxylic acids, esters, amides. Elements of nomenclature.

# Mathematics

- **Number sets and algebra:**

natural, integer, rational, real numbers. Ordering and comparison; order of magnitude and scientific notation. Operations and their properties. Proportions and percentages. Powers (integer, rational exponents) and their properties. Radicals and their properties. Logarithms (base 10 and base e) and their properties. Elements of combinatorics. Algebraic expressions, polynomials. Special products, nth power of a binomial, factoring polynomials. Algebraic fractions. First- and second-degree equations and inequalities. Systems of equations.

- **Functions:**

fundamental notions and graphical representations (domain, codomain, sign study, continuity, maxima/minima, increasing/decreasing behavior). Elementary functions: algebraic (whole and fractional), exponential, logarithmic, trigonometric. Composite and inverse functions. Trigonometric equations and inequalities.

- **Geometry:**

polygons and their properties. Circumference and circle. Measurement of lengths, areas, volumes. Isometries, similarities, equivalences in the plane. Geometric loci. Angle measurement in degrees and radians. Sine, cosine, tangent and their notable values. Trigonometric formulas. Solving triangles. Cartesian coordinate system. Distance between two points, midpoint of a segment. Equation of a line. Parallelism and perpendicularity. Distance of a point from a line. Equations of the circle, parabola, hyperbola, ellipse. Pythagorean theorem. Euclid's theorems (first and second).

- **Probability and statistics:**

frequency distributions and main graphical representations. Random experiments and events. Probability and frequency.

# Physics

- **Physical quantities and measurement:**

fundamental and derived quantities. Unit systems (International, Technical). Multiples and submultiples. Scientific notation. Unit conversions. Scalar and vector quantities. Vectors and vector operations.

- **Kinematics:**

describing motion. Velocity and angular velocity, acceleration and centripetal acceleration. Uniform rectilinear motion, uniformly accelerated motion, uniform circular motion, simple harmonic motion.

- **Dynamics:**

force as an interaction between bodies. Forces as applied vectors. The principle of inertia. Mass and Newton's 2nd law. Weight, elastic force, static/kinetic friction. Action and reaction (3rd law). Impulse and momentum, conservation of momentum. Moment of a force and angular momentum. Work and kinetic energy. Conservative forces and potential energy. Conservation of mechanical energy. Power.

- **Fluid mechanics:**

density and compressibility. Hydrostatics: pressure, Pascal's, Stevin's, and Archimedes' principles. Fluid dynamics: flow rate, continuity equation. Ideal fluids and Bernoulli's equation. Viscous forces in real fluids.

- **Thermodynamics:**

equilibrium, temperature, thermometers. Heat and calorimetry. Modes of heat propagation. Heat capacity and specific heat. Changes of state, latent heat. Laws of ideal gases. First and second laws of thermodynamics.

- **Electricity & electromagnetism:**

electric charges, Coulomb's law. Electric field and potential, equipotential surfaces. Dielectric constant, capacitance, capacitors. Electrostatic energy. Series/parallel capacitors. Generators, voltage, current. Resistivity, resistance, Ohm's law. Series/parallel resistors. Kirchhoff's laws. Work, power, Joule effect. DC/AC current, period and frequency. Magnetic field of a current. Forces on currents in a magnetic field. Electromagnetic induction.



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